

Multiple Choice (10 marks)

1. A rise in intracellular free calcium in the sea urchin oocyte causes the release of proteolytic enzymes which act to prevent polyspermy. The events just described entail the
 - (a) polyspermy reaction
 - (b) enzymatic reaction
 - (c) proteolytic reaction
 - (d) acrosomal reaction
 - (e) oocyte reaction
2. All of the following play a role in morphogenesis EXCEPT
 - (a) homeotic genes
 - (b) cell division
 - (c) signal transduction
 - (d) cell differentiation
 - (e) apoptosis
3. From 146 students tested for PTC tasting ability, 105 were tasters and 41 were non-tasters. Calculate the frequencies of tasters and non-tasters.
 - (a) Frequency of T is 0.59 and frequency of t is 0.41
 - (b) Frequency of T is 0.47 and frequency of t is 0.53
 - (c) Frequency of T is 0.72 and frequency of t is 0.28
 - (d) Frequency of T is 0.50 and frequency of t is 0.50
 - (e) Frequency of T is 0.41 and frequency of t is 0.59
4. What prevents the stomach from being digested by its own secretions?
 - (a) The thick muscular wall of the stomach
 - (b) Antibacterial properties of stomach acid
 - (c) Gastric juice
 - (d) High pH levels in the stomach
 - (e) Presence of beneficial bacteria
5. When oxygen becomes unavailable, this process regenerates NAD^+ , allowing respiration to continue.

- (a) Calvin cycle
- (b) Fermentation
- (c) Glycolysis
- (d) Oxidative phosphorylation
- (e) Pyruvate oxidation

True / False (10 marks)

Answer True or False

- 6. True or False: If mRNA molecules are not degraded, they would replicate rapidly, leading to uncontrolled protein synthesis and cellular dysfunction.
- 7. True or False: Smaller cells tend to have a larger number of organelles compared to larger cells, enhancing their functional capacity.
- 8. True or False: Hemoglobin's affinity for O₂ increases in response to elevated H⁺ concentration in the blood.
- 9. True or False: Oxygen is a principal inorganic compound essential for life, participating in cellular respiration and energy production.
- 10. True or False: An increase in the amount of CO₂ in the blood has no effect on the regulation of respiration in humans.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss how specific floral adaptations, such as pendant red-colored flowers, can influence pollination dynamics by limiting bee access while promoting hummingbird visitation.
- 12. Discuss the reasons why a population of paramecia, originating from a single individual through fission, would not engage in conjugation. Include an analysis of genetic identity and its implications.

End of Paper